

Unit 202 — Semantic Reference QA Remediation Review

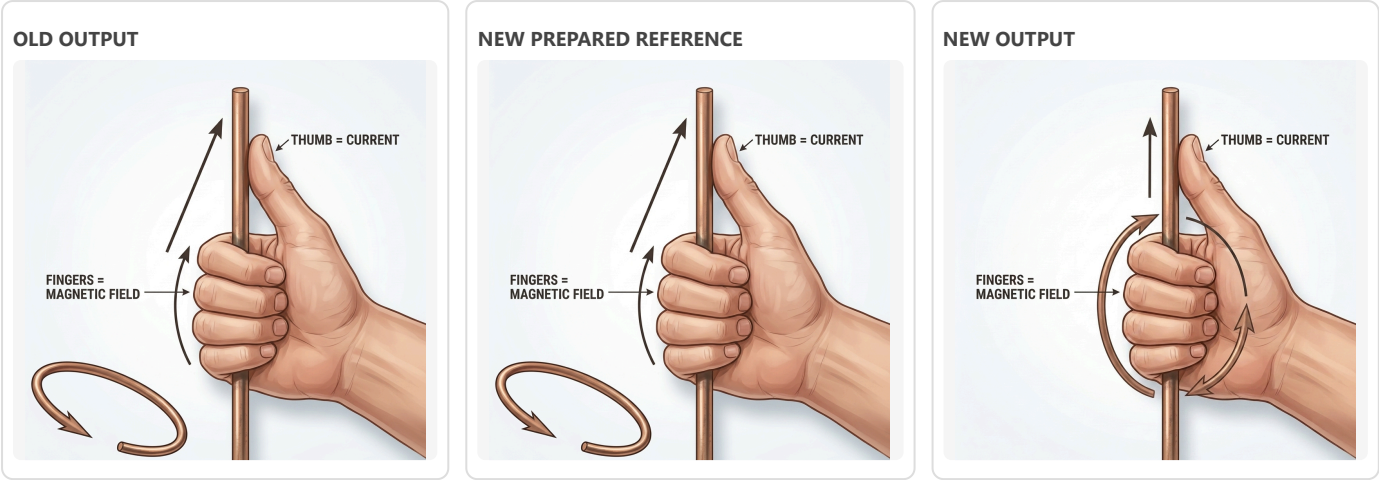
CC-11.12 · Generated: 2026-08-25T23:36:30.051Z

UNIT 202 GENERATIVE VISUAL REMEDIATION COMPLETE — AWAITING PRODUCT OWNER / CHATGPT REVIEW

REDO entries remediated	22
Overall PASS	22
HUMAN_REVIEW_REQUIRED	0
Missing prepared reference (hard failure)	0
TECHNICAL verdict PASS	22 / 22
PEDAGOGICAL_CLARITY verdict PASS	22 / 22
VISUAL_PRODUCT_QUALITY verdict PASS	22 / 22

Semantic finding: Handedness is correct, but the field-direction cue does not visually teach curled fingers = field circulation.

Required action: Keep right-hand reference but require the magnetic-field arrow to follow the curl of the fingers around the conductor; remove detached little-finger-to-index-finger arrow.

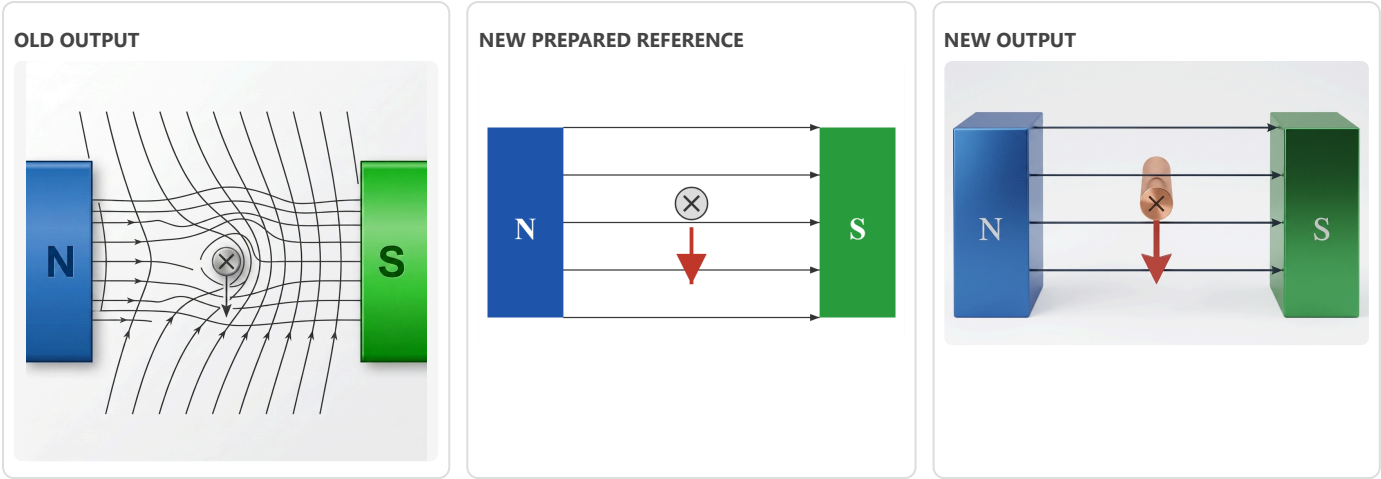


Keep elements	right hand; thumb = conventional current direction; conductor
Crop/remove elements	the detached little-finger-to-index-finger arrow (reads as an unrelated hand-internal gesture, not field circulation)
Required final elements	a field-direction arrow that follows the curl of the fingers around the conductor (a curved arrow tracing the finger wrap, not a straight or detached stroke)
Prohibited final elements	any arrow disconnected from the finger curl; left hand; thumb pointing away from conventional current
Model must not infer	which way the fingers curl -- must match the already-correct right-hand grip geometry, not be redrawn ambiguously
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Both named defects (detached arrow icon, non-curl-tracing field arrow) resolved.

Semantic finding: Pedagogically opaque; competing/duplicated field/force cues.

Required action: Build exact state reference: one N/S pair, one N→S magnetic-field-line system, X/dot conductor marker, one reference-authorized force arrow. No direction re-derivation.

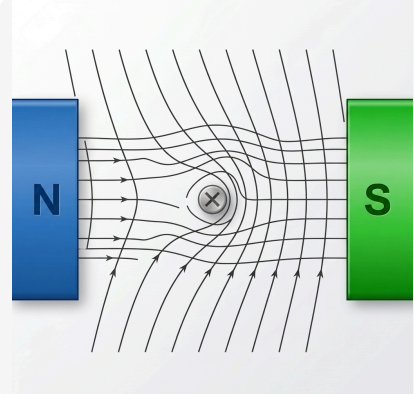


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged horizontally (side by side)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker (X for into-page, dot for out-of-page); ONE force arrow, already correct in the prepared reference board (never re-derived during generation)
Prohibited final elements	a second/competing field or force line system; any force arrow not already authorised by the prepared board
Model must not infer	force direction -- must come from the prepared, reference-authorized board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

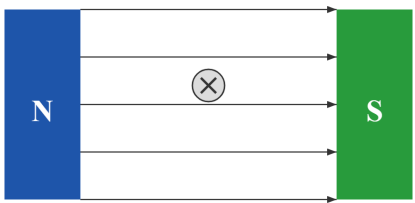
PASS -- semantic QA REDO resolved. Clean single field-line system, correct X marker, correct downward force arrow.

Semantic finding: Assessment sibling should differ only by answer cue.
Required action: Same exact apparatus as teaching sibling but withhold force arrow only.

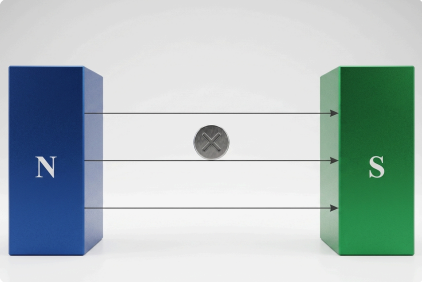
OLD OUTPUT



NEW PREPARED REFERENCE



NEW OUTPUT

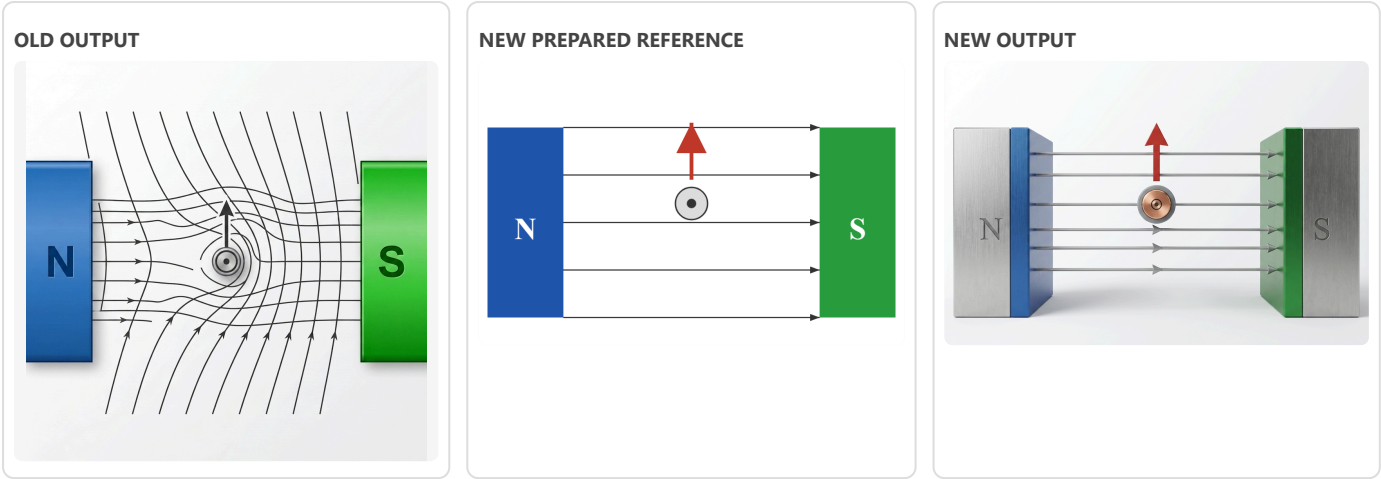


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged horizontally (side by side)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker matching the teaching sibling (the given stimulus)
Prohibited final elements	any force arrow or other reveal of the assessed force direction; text such as 'Force' unless used consistently across the whole family
Model must not infer	force direction -- must come from the prepared, reference-authorised board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Given stimulus (X) shown, force withheld.

Semantic finding: Pedagogically opaque; competing/duplicated field/force cues.

Required action: Build exact state reference: one N/S pair, one N→S magnetic-field-line system, X/dot conductor marker, one reference-authorized force arrow. No direction re-derivation.

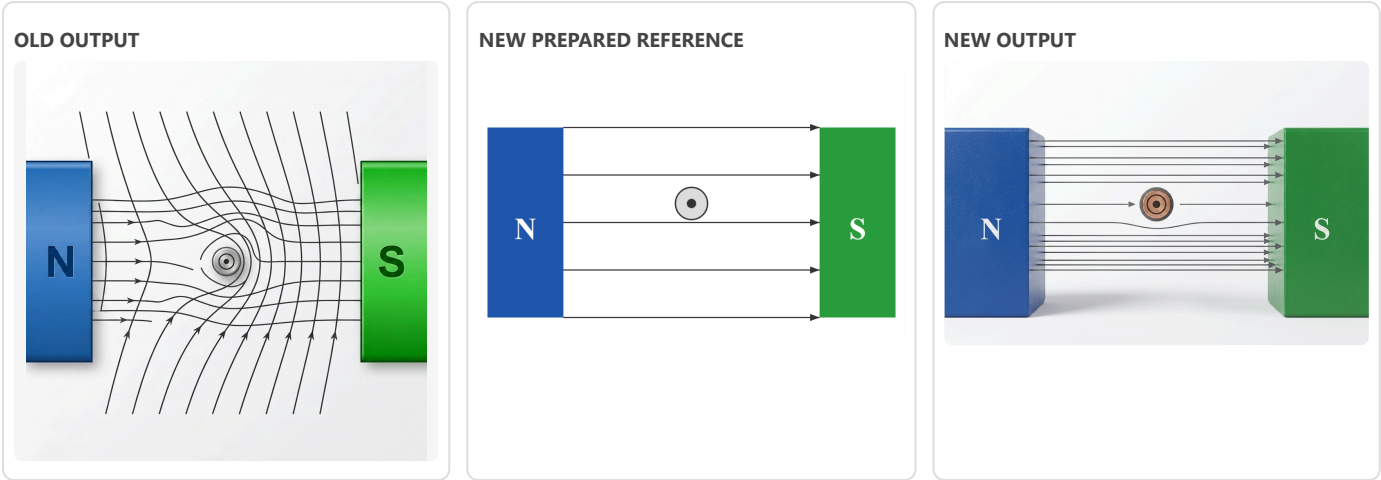


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged horizontally (side by side)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker (X for into-page, dot for out-of-page); ONE force arrow, already correct in the prepared reference board (never re-derived during generation)
Prohibited final elements	a second/competing field or force line system; any force arrow not already authorised by the prepared board
Model must not infer	force direction -- must come from the prepared, reference-authorized board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Clean field system, correct dot marker, correct upward force arrow.

Semantic finding: Assessment sibling should differ only by answer cue.

Required action: Same exact apparatus as teaching sibling but withhold force arrow only.

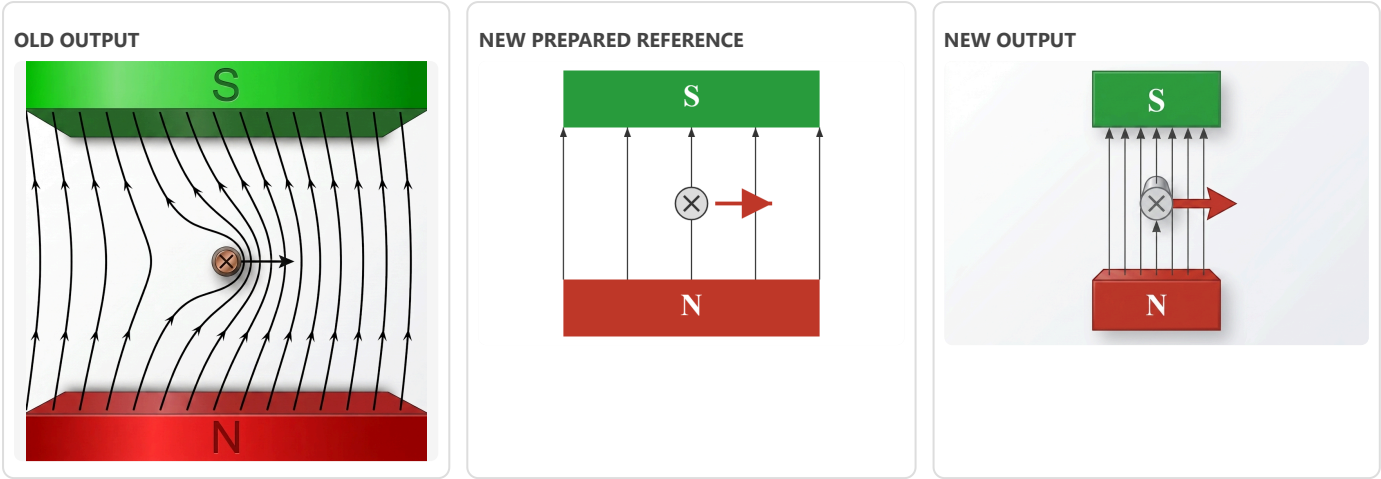


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged horizontally (side by side)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker matching the teaching sibling (the given stimulus)
Prohibited final elements	any force arrow or other reveal of the assessed force direction; text such as 'Force' unless used consistently across the whole family
Model must not infer	force direction -- must come from the prepared, reference-authorised board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Given stimulus (dot) shown, force withheld. Field-line density slightly higher than ideal but all lines consistent direction, no tangling/contradiction -- accepted.

Semantic finding: Pedagogically opaque; competing/duplicated field/force cues.

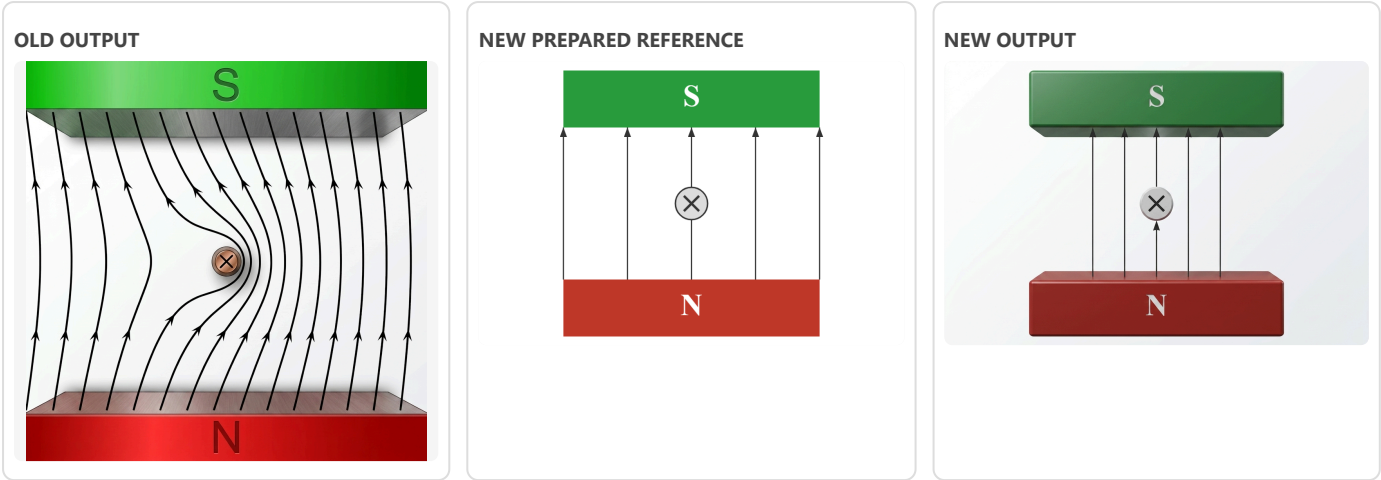
Required action: Build exact state reference: one N/S pair, one N→S magnetic-field-line system, X/dot conductor marker, one reference-authorized force arrow. No direction re-derivation.



Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged vertically (stacked)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker (X for into-page, dot for out-of-page); ONE force arrow, already correct in the prepared reference board (never re-derived during generation)
Prohibited final elements	a second/competing field or force line system; any force arrow not already authorised by the prepared board
Model must not infer	force direction -- must come from the prepared, reference-authorized board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Clean field system, correct X marker, correct rightward force arrow.

Semantic finding: Assessment sibling should differ only by answer cue.
Required action: Same exact apparatus as teaching sibling but withhold force arrow only.

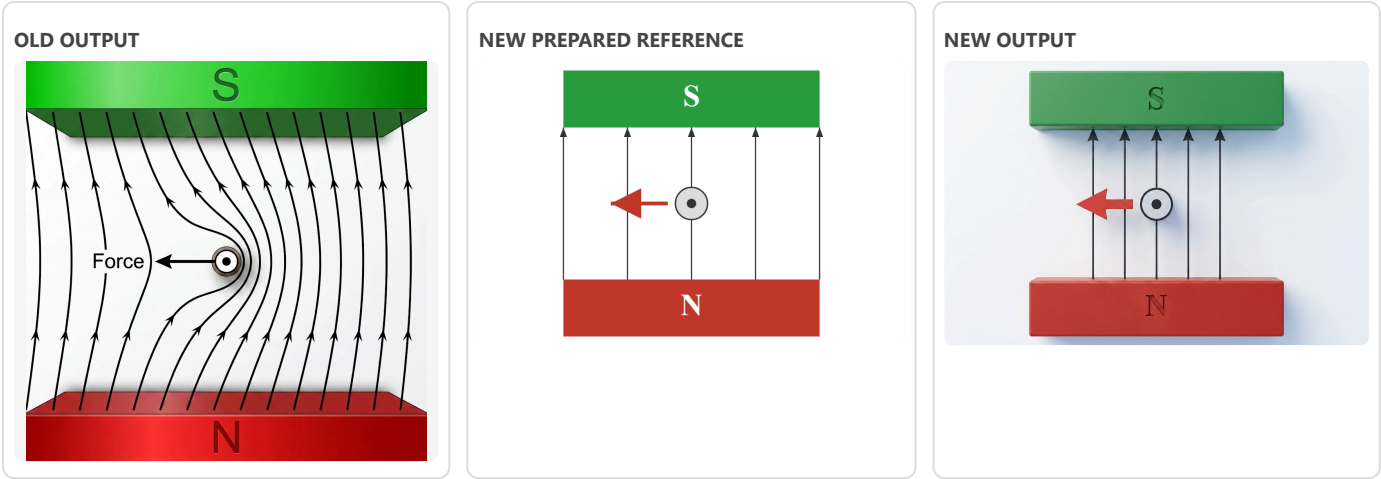


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged vertically (stacked)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker matching the teaching sibling (the given stimulus)
Prohibited final elements	any force arrow or other reveal of the assessed force direction; text such as 'Force' unless used consistently across the whole family
Model must not infer	force direction -- must come from the prepared, reference-authorised board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Given stimulus (X) shown, force withheld.

Semantic finding: Pedagogically opaque; competing/duplicated field/force cues.

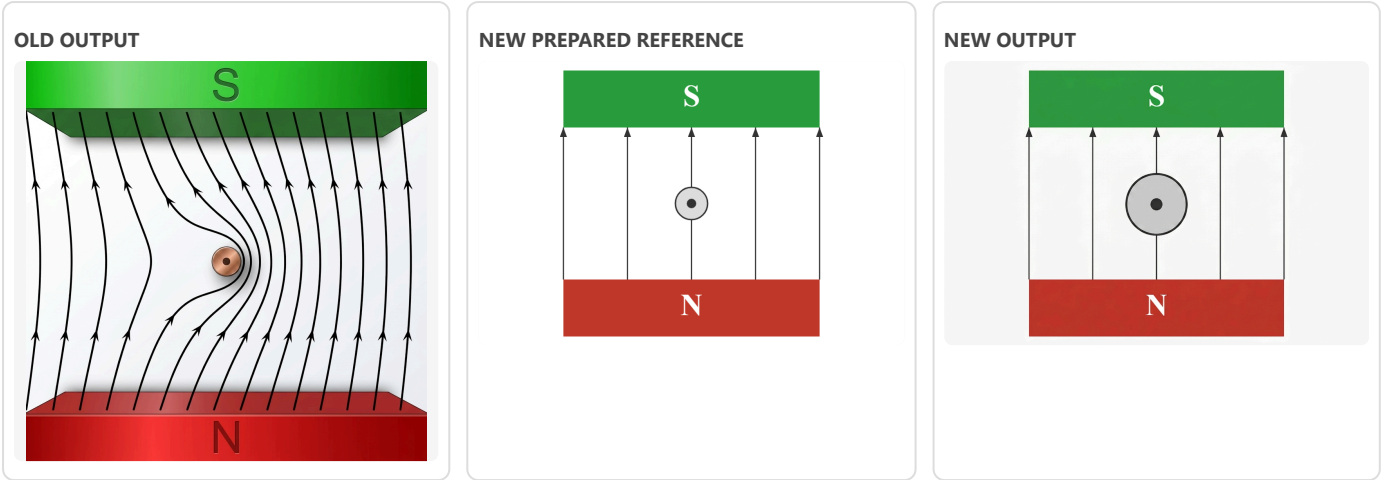
Required action: Build exact state reference: one N/S pair, one N→S magnetic-field-line system, X/dot conductor marker, one reference-authorized force arrow. No direction re-derivation.



Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged vertically (stacked)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker (X for into-page, dot for out-of-page); ONE force arrow, already correct in the prepared reference board (never re-derived during generation)
Prohibited final elements	a second/competing field or force line system; any force arrow not already authorised by the prepared board
Model must not infer	force direction -- must come from the prepared, reference-authorized board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS -- semantic QA REDO resolved. Clean field system, correct dot marker, correct leftward force arrow.

Semantic finding: Assessment sibling should differ only by answer cue.
Required action: Same exact apparatus as teaching sibling but withhold force arrow only.

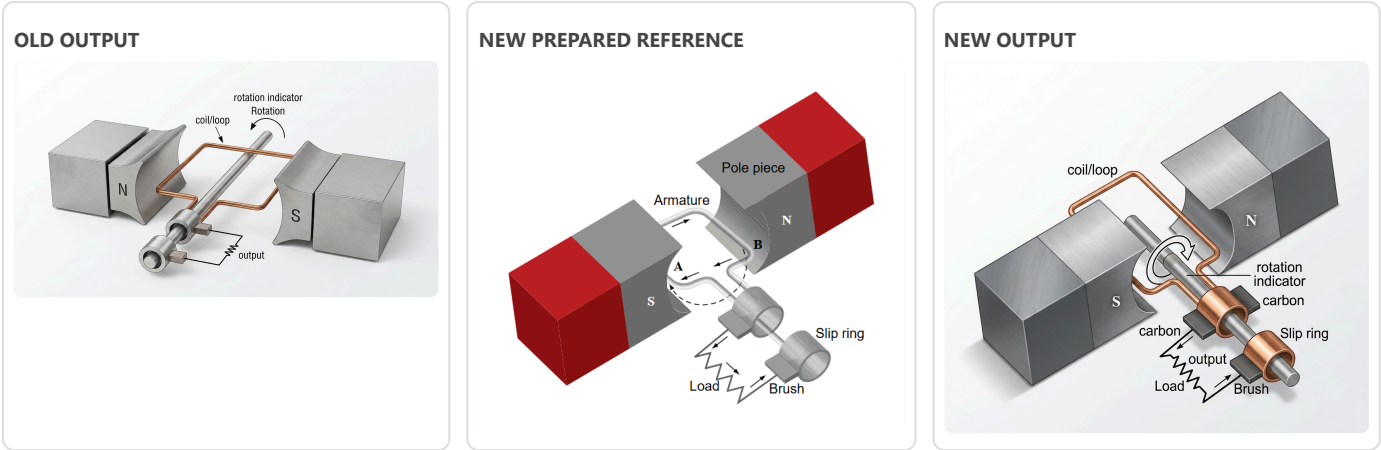


Keep elements	ONE N/S pole pair; ONE N→S magnetic-field-line system (N/S poles arranged vertically (stacked)); ONE X/dot current marker at the conductor
Crop/remove elements	any competing/duplicated second field-line or force-line system
Required final elements	ONE N/S pair; ONE field-line system; ONE current marker matching the teaching sibling (the given stimulus)
Prohibited final elements	any force arrow or other reveal of the assessed force direction; text such as 'Force' unless used consistently across the whole family
Model must not infer	force direction -- must come from the prepared, reference-authorised board only; Claude/Gemini must never independently re-derive it during generation
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS after 1 bounded correction -- attempt 2 rendered the conductor as a 3D copper cylinder with no clear dot marker (ambiguous current-direction stimulus); corrected to a clear flat circular dot marker matching its sibling's style. Field-line density also normalised to 5 (attempt 2 had ~10).

Semantic finding: Review says reference preview not found; generated topology is visually ambiguous and can read as two loops/coils.

Required action: Use an actual visible exact reference. Preferred: FAA-derived public-domain Elementary generator.svg or explicit DOE Figure 1 crop. Prepare simplified face-on frame: ONE loop, shaft, two slip rings, two brushes, N/S, output.



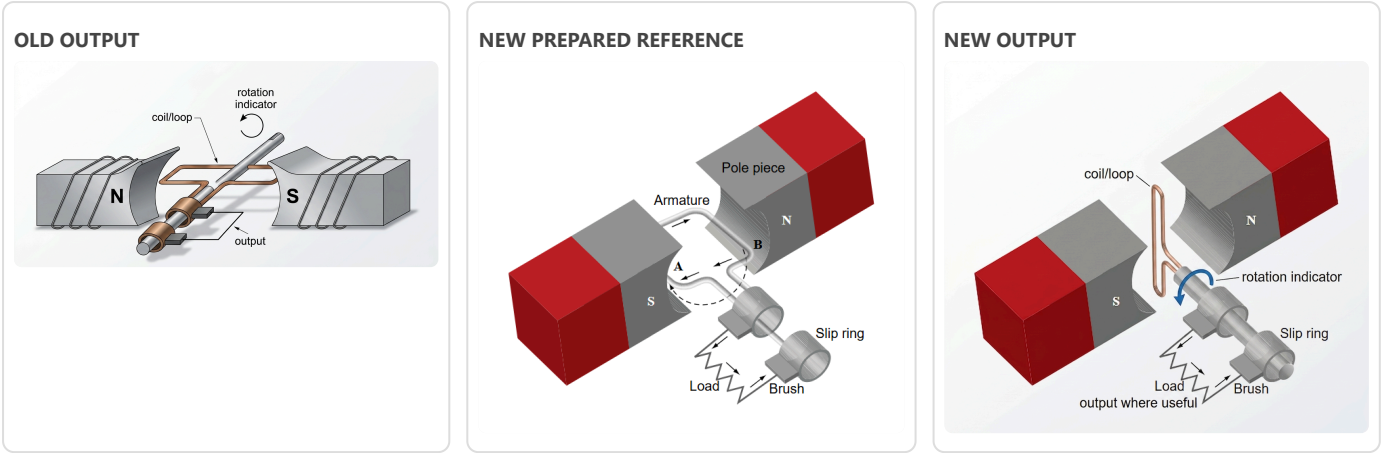
Keep elements	one N/S pole pair; one shaft; two separate slip rings, concentric on the shaft; two stationary brushes; one rectangular loop; output leads from the brushes
Crop/remove elements	any second loop/coil-like structure; any part of the reference beyond the elementary single-loop generator
Required final elements	exactly ONE unmistakable rectangular loop (must not be readable as two coils); one shaft; two slip rings; two brushes; N/S pole labels; a coherent electrical path from loop to output
Prohibited final elements	a second/duplicate coil; a split-ring commutator (that is a DC motor, not this AC generator); floating/disconnected rings; decorative output wiring with no coherent path
Model must not infer	loop/shaft/slip-ring topology -- must be preserved exactly from the prepared reference, never reconstructed from memory or prose
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. The core semantic QA defect (reference never actually visible, topology ambiguous) is resolved by using a real, visible, unambiguous reference directly.

Simple rotating-loop AC generator — loop plane edge-on to poles — near-peak EMF

unit202.generator.rotating-loop.vertical

Semantic finding: Current output is too complicated.
Required action: Second exact frame from the same one-loop topology, rotated to edge-on/near-peak state. Must visibly be the same one loop/shaft/two-ring system.



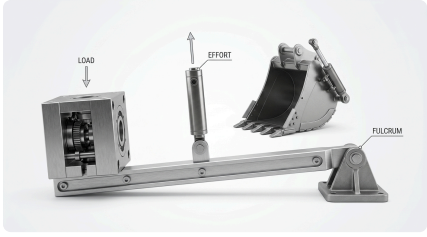
Keep elements	one N/S pole pair; one shaft; two separate slip rings, concentric on the shaft; two stationary brushes; one rectangular loop; output leads from the brushes
Crop/remove elements	any second loop/coil-like structure
Required final elements	exactly ONE loop, visibly the SAME apparatus as the horizontal sibling, rotated to edge-on; two slip rings; two brushes; N/S; coherent output path
Prohibited final elements	a second/duplicate coil; a split-ring commutator; an apparatus that looks like a different machine from its horizontal sibling
Model must not infer	the edge-on rotation -- the prepared reference frame itself must already show this pose, not be derived by the model rotating the face-on pose in its head
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Clearly the same one-loop system as the horizontal sibling, correctly rotated to the edge-on pose.

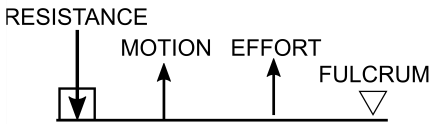
Semantic finding: Digger cue contaminated generation into gearbox/actuator/bucket.

Required action: Crop ONLY abstract Class-III schematic from PSF source, excluding excavator/digger, or use direct ThirdClassLever.svg. Match Class I/II family.

OLD OUTPUT



NEW PREPARED REFERENCE



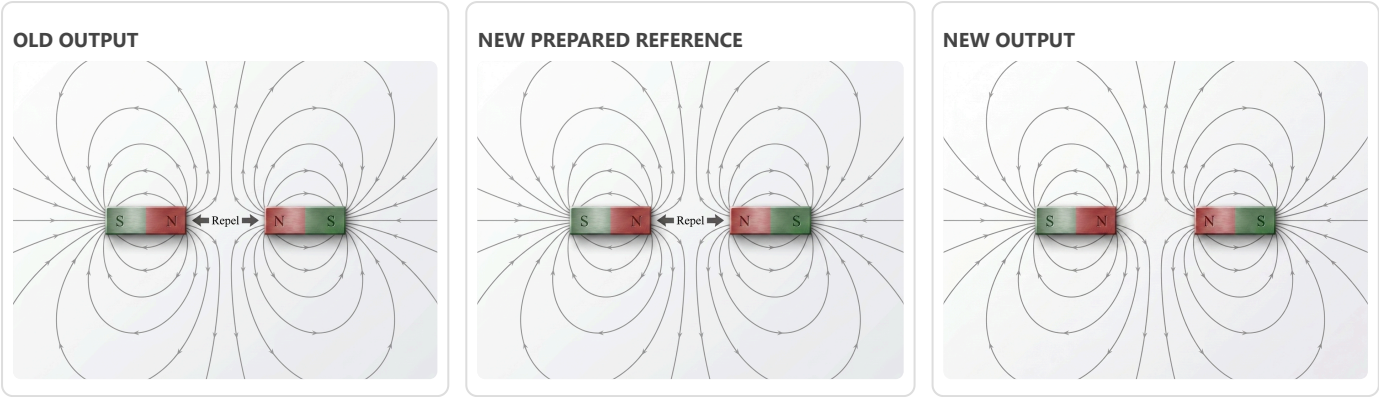
NEW OUTPUT



Keep elements	abstract Class-III beam/fulcrum/effort/load arrangement from the PSF source (or ThirdClassLever.svg directly)
Crop/remove elements	the excavator/digger real-world example -- must never be passed to Gemini at all, not even as background context
Required final elements	ONE Class III arrangement only (fulcrum at one end, effort between fulcrum and load, load at the other end); EFFORT / LOAD / FULCRUM labels; premium metallic beam finish matching Class I/II
Prohibited final elements	any excavator, digger, bucket, hydraulic-arm or other real-world machine iconography; Class I or Class II geometry mixed in
Model must not infer	
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. The core semantic QA defect (excavator contamination) is fully resolved. Geometry is correct Class III. Minor extra-label cosmetic imperfection accepted after one correction attempt made things worse, not better.

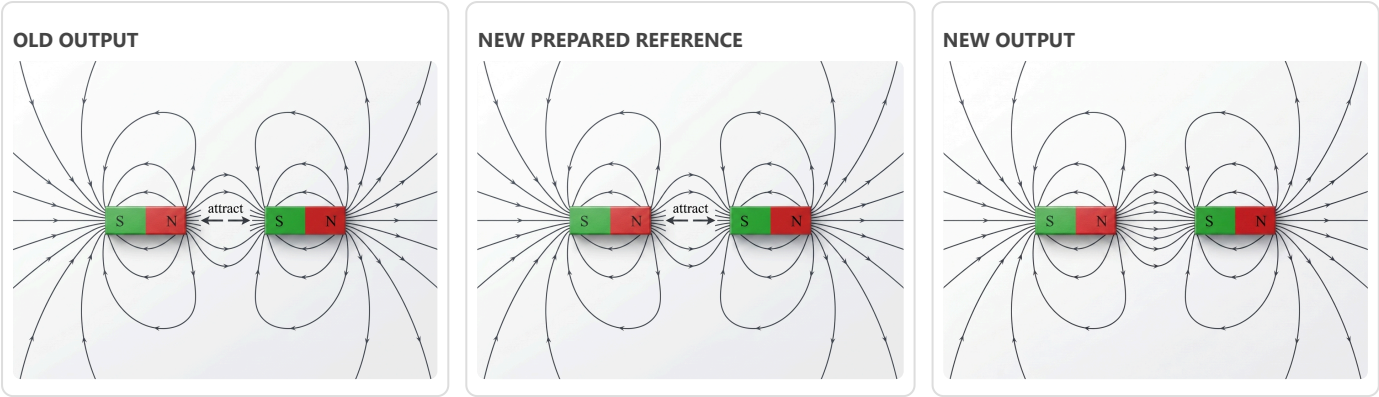
Semantic finding: Current output is fuzzy and adds unnecessary answer-like text.
Required action: Clean/regenerate: no Repel word, no explanatory force label; field geometry and N/S should communicate repulsion; improve sharpness.



Keep elements	two bar magnets, like poles facing; N/S labels; compressed/deflected field-line geometry between the magnets
Crop/remove elements	the word 'Repel'; any explanatory force label/caption
Required final elements	sharp, crisp field-line rendering (prior output was fuzzy); N/S labels on both magnets; field geometry that itself communicates repulsion (visible gap, deflected/compressed lines)
Prohibited final elements	the word 'Repel' or any textual restatement of the answer; force-arrow callout text
Model must not infer	
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Both named defects (Repel text, fuzzy rendering) resolved.

Semantic finding: Field pattern should communicate attraction without baked answer words.
Required action: Clean/regenerate: remove Attract and double-headed arrow; retain N/S and physically meaningful connecting field lines.

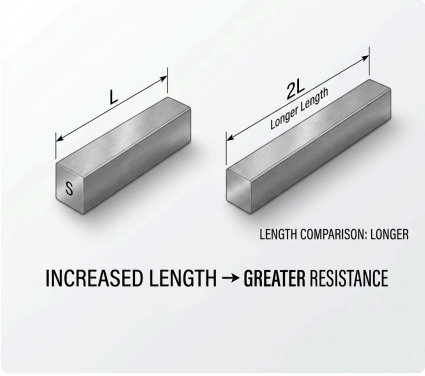


Keep elements	two bar magnets, unlike poles facing; N/S labels; merging field-line geometry between the magnets
Crop/remove elements	the word 'Attract'; the double-headed attraction arrow
Required final elements	N/S labels on both magnets; physically meaningful connecting field lines (merging between the magnets, consistent with attraction)
Prohibited final elements	the word 'Attract'; any double-headed arrow or other explicit force-direction callout
Model must not infer	
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Named defect (attract text + arrow) resolved.

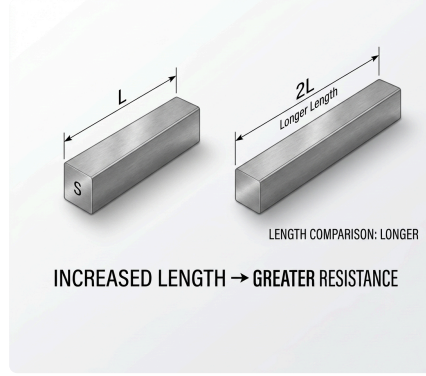
Semantic finding: Source notation S conflicts with course $R=\rho L/A$.
Required action: Prepare controlled board using A for cross-sectional area or no area symbol. Remove inherited S. Same material/A, vary only L; avoid baked conclusion if reuse/assessment needs neutrality.

OLD OUTPUT



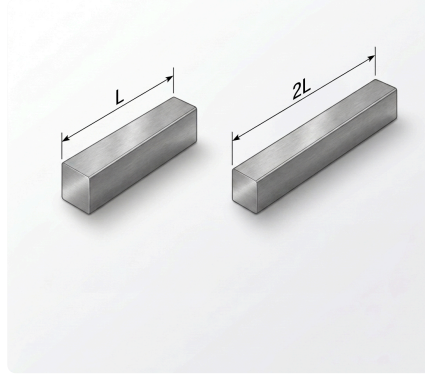
INCREASED LENGTH → GREATER RESISTANCE

NEW PREPARED REFERENCE



INCREASED LENGTH → GREATER RESISTANCE

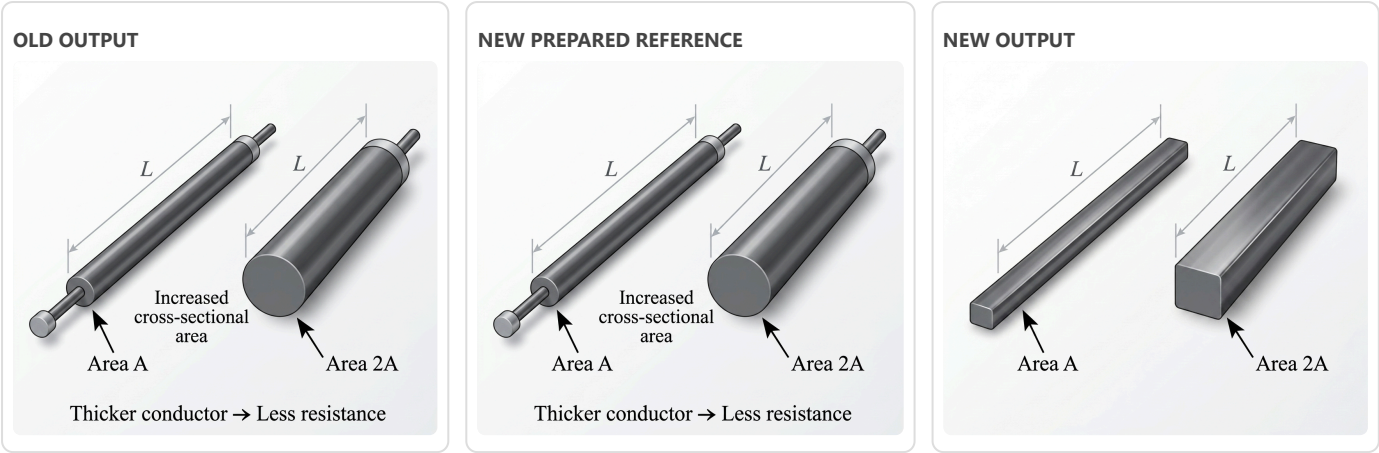
NEW OUTPUT



Keep elements	same-material wire comparison composition; clear length difference between the two samples
Crop/remove elements	the inherited 'S' cross-sectional-area notation from the source geometry reference
Required final elements	length labelled L (or L / 2L, etc.); cross-sectional area labelled A if labelled at all -- never S; same material/finish on both samples; only length varies
Prohibited final elements	the symbol S for area or any other course-inconsistent notation; a second varying variable (area or material must stay fixed)
Model must not infer	which variable is fixed vs varied -- both samples must be visibly the same material and area, only length different
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Notation conflict and baked conclusion both resolved.

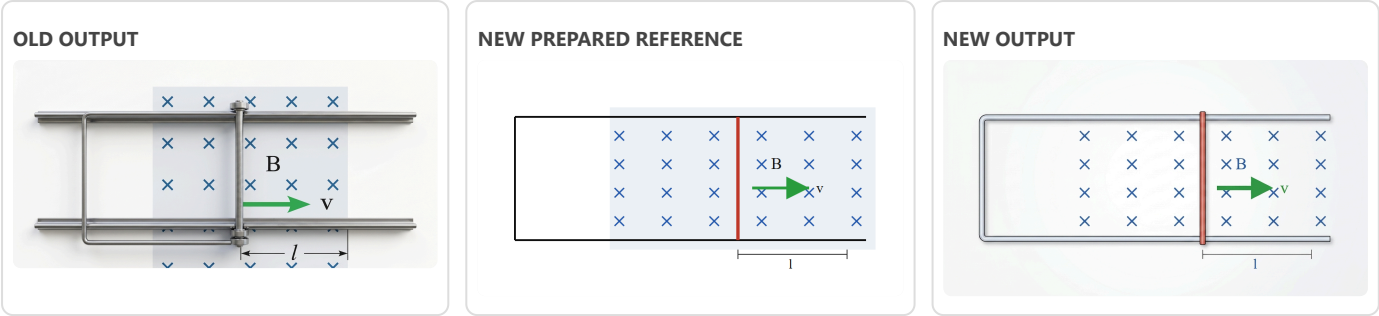
Semantic finding: Family should use one governed notation/grammar.
Required action: Rebuild as matched sibling: same material/L, vary only A; use A/2A consistently; minimal labels.



Keep elements	same-material wire comparison composition; clear area difference between the two samples
Crop/remove elements	the inherited 'S' cross-sectional-area notation from the source geometry reference
Required final elements	cross-sectional area labelled A / 2A consistently; length labelled L if labelled at all, held fixed; same material/finish on both samples; only area varies
Prohibited final elements	the symbol S for area; a second varying variable (length or material must stay fixed)
Model must not infer	which variable is fixed vs varied -- both samples must be visibly the same material and length, only area different
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Baked conclusion removed; family style now matches its sibling.

Semantic finding: Current pseudo-3D output destroys clarity.
Required action: Minimal verified 2D reference: one moving conductor, rails only if needed, one B convention, one v arrow, one l dimension. Forbid photoreal/3D pipe-like apparatus.



Keep elements	one moving conductor; B-field convention (dot/cross or arrows); v arrow; l dimension
Crop/remove elements	photoreal pipe-like / 3D-surface apparatus rendering from any prior attempt
Required final elements	one conductor (rails only if strictly needed for the circuit); one B convention; one v arrow; one l dimension label
Prohibited final elements	photoreal 3D pipes, tubes, or apparatus casings; any second competing conductor/rail system not needed for the minimal circuit
Model must not infer	the B/v/l geometric relationship -- must be preserved from the authoritative motional-EMF panel, not re-imagined as a 3D scene
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Photoreal 3D clutter removed; minimal 2D clarity achieved.

Semantic finding: Capacitor breaks family consistency if it alone includes the symbol.
Required action: Keep physical capacitor reference but remove UK/IEC symbol from physical-recognition raster if present. Symbol remains separate deterministic card.

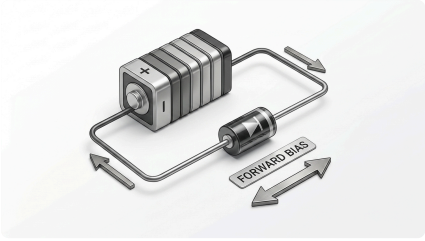


Keep elements	existing approved physical capacitor reference/geometry
Crop/remove elements	the UK/IEC circuit symbol, if present anywhere in the physical-recognition raster
Required final elements	physical capacitor component + name only
Prohibited final elements	any schematic/circuit symbol baked into this raster
Model must not infer	
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

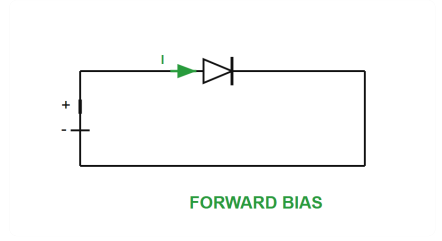
PASS. Family-consistency defect (symbol only on this one asset) resolved.

Semantic finding: Current image is hard to understand; semiconductor-band reference is the wrong composition authority.
Required action: Build exact Level-2 reference using governed diode symbol, simple DC source and conventional-current arrow. Same composition as reverse sibling.

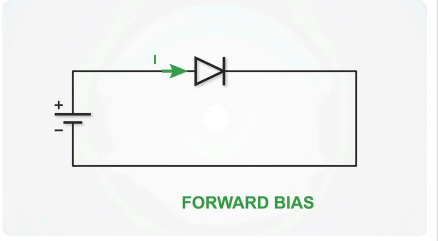
OLD OUTPUT



NEW PREPARED REFERENCE



NEW OUTPUT

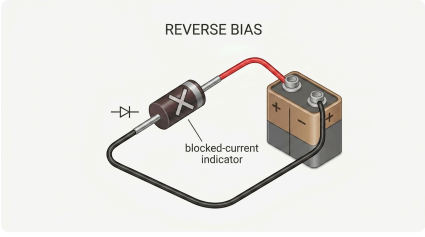


Keep elements	governed diode schematic symbol; simple DC source; conventional-current arrow
Crop/remove elements	any semiconductor-band-diagram composition authority from a prior attempt
Required final elements	diode symbol oriented for forward bias; source polarity matching forward bias (anode side more positive); visible current-flow indicator; FORWARD BIAS label
Prohibited final elements	semiconductor depletion-band/energy-band imagery as the composition basis; any current indicator suggesting current is blocked
Model must not infer	which terminal is anode/cathode or which way current flows -- the prepared circuit reference must already encode this exactly
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

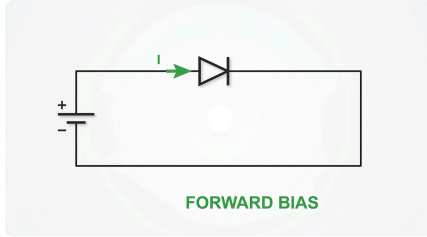
PASS. Wrong-composition-authority defect resolved.

Semantic finding: Forward/reverse should be a one-variable comparison.
Required action: Use same circuit as forward sibling; reverse only source polarity/state and show blocked current clearly.

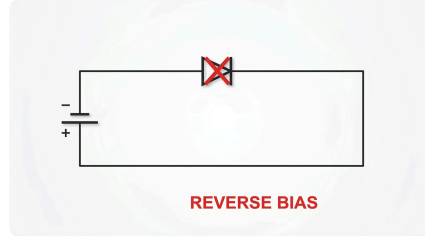
OLD OUTPUT



NEW PREPARED REFERENCE



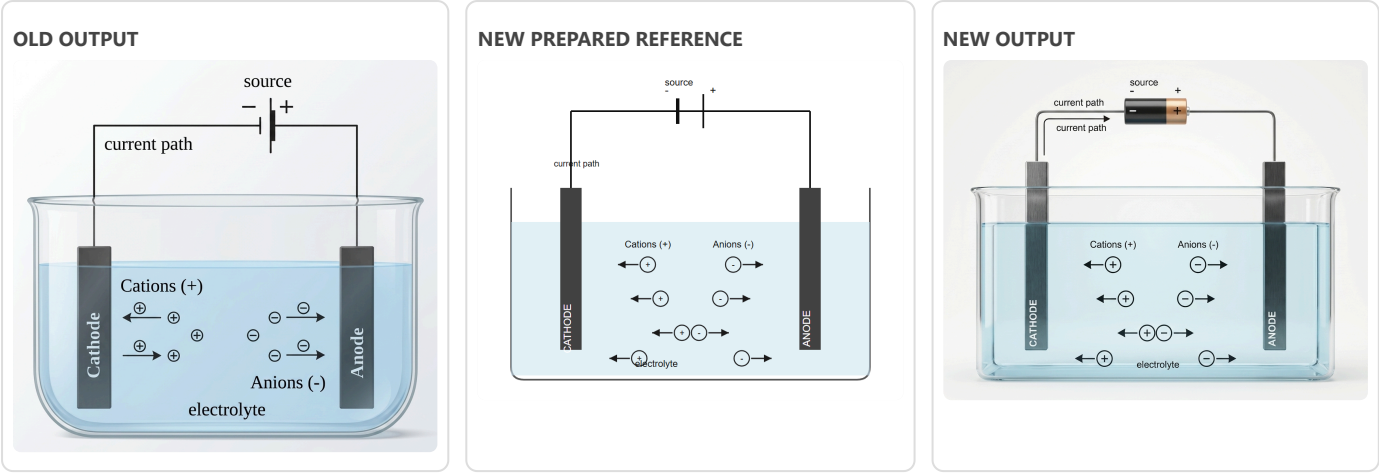
NEW OUTPUT



Keep elements	governed diode schematic symbol; simple DC source; same circuit layout as the forward sibling
Crop/remove elements	any semiconductor-band-diagram composition authority from a prior attempt
Required final elements	diode symbol oriented for reverse bias; source polarity matching reverse bias (cathode side more positive); current clearly shown as blocked (e.g. a blocked/X indicator); REVERSE BIAS label
Prohibited final elements	semiconductor depletion-band/energy-band imagery as the composition basis; any indicator suggesting current flows
Model must not infer	which terminal is anode/cathode or the blocked-current state -- the prepared circuit reference must already encode this exactly
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. Family-consistency and polarity-ambiguity defects from attempt 3 both resolved.

Semantic finding: At least one cation-side arrow is visibly wrong despite automated PASS.
Required action: Prepare exact board with battery, electrodes, electrolyte, cations/anions and EVERY ion arrow already correct. Cations→cathode; anions→anode. No meters. Gemini may restyle but not add/remove/reverse arrows.



Keep elements	battery/source with +/- terminals; cathode and anode electrodes; electrolyte vessel; current-path wiring
Crop/remove elements	voltmeter, ammeter, or any other measurement instrument if reintroduced
Required final elements	battery; electrodes (labelled cathode/anode); electrolyte; EVERY ion arrow already correct in the prepared board: all cations -> cathode, all anions -> anode
Prohibited final elements	voltmeter; ammeter; any bench power-supply box in addition to the simple battery symbol; any invented wire directly shorting the two electrodes
Model must not infer	ion-migration direction -- every arrow must already be correct in the prepared board; Gemini may restyle rendering but must not add, remove, or reverse any arrow
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. The specific micro-directional defect the semantic QA found (wrong cation arrow) is resolved and independently re-verified pixel-by-pixel, not merely re-trusted.

Semantic finding: Two near-identical stripped cables do not teach the conceptual comparison.

Required action: Retire Stranded lamp wire.jpg as primary teaching reference. Build purpose-specific cable cutaway or controlled metal-vs-polymer comparison with unmistakable conductor/insulator roles.



Keep elements	conductive metal core; insulating polymer sheath
Crop/remove elements	'Stranded lamp wire.jpg' retired as the primary composition reference (two near-identical stripped cables do not teach the comparison)
Required final elements	an unmistakable conductor role (bare/exposed conductive metal, clearly current-carrying); an unmistakable insulator role (clearly non-conductive polymer/sheath material); purpose-built cutaway or controlled metal-vs-polymer comparison composition
Prohibited final elements	two visually near-identical stripped cable ends as the sole comparison
Model must not infer	
TECHNICAL	PASS
PEDAGOGICAL_CLARITY	PASS
VISUAL_PRODUCT_QUALITY	PASS
OVERALL	PASS

PASS. The pedagogical-weakness defect (two identical cables, no clear comparison) is resolved.